

Misattribution Machine: AI-Driven Detection of Archival Misattribution

Technical Documentation — Exhibition System

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1. Project Overview

The Misattribution Machine is an exhibition system designed to detect and visualise the propagation of misattribution errors across digital archives and AI language models. The project analyses how incorrect nationality labels assigned to five artists — Kazimir Malevich, Alexandra Exter, Vera Pagava, David Kakabadze, and Sergei Parajanov — originate in archival authority records and are subsequently reproduced by major AI systems including Claude, GPT-4o, and LLAMA.

The system comprises three principal components:

- (1) A dataset analysis pipeline processing 659 archival records;
- (2) A real-time AI misattribution detection tool using the Claude API; and
- (3) A 12-screen LED exhibition installation rendered in p5.js.

2. Dataset Analysis

Two datasets were processed to characterise the nature and scale of misattribution across institutional records.

2.1 Parajanov Bibliographic Dataset

305 records were analysed from bibliographic sources relating to Sergei Parajanov. The misattribution breakdown is as follows:

Misattribution Type	Record Count	Percentage
Wrong nationality label	218	71.5%
Omission of national identity	54	17.7%
Spelling / name variants	33	10.8%
Total	305	100%

2.2 Artworks Dataset (Malevich, Exter, Pagava, Kakabadze)

354 records were analysed across the four visual artists. The breakdown by artist and misattribution type is presented below:

Artist	Total Records	Wrong Label	Omission	Spelling / Structural
Kazimir Malevich	272	195	61	16
Alexandra Exter	44	38	4	2
Vera Pagava	19	19	—	—
David Kakabadze	19	—	—	19 (structural)
Subtotal	354	252	65	37

2.3 Combined Dataset Summary

Metric	Value
Total records	659
Artists covered	5
Records containing at least one misattribution	~94%

2.4 Artist Profiles and Misattribution Context

Kazimir Malevich. Born in Kyiv, 1879. Routinely labelled 'Russian avant-garde' or 'Soviet artist' in major archives including the Library of Congress, WorldCat, and MoMA, erasing his Ukrainian origin.

Alexandra Exter. Born in Kyiv; trained at the Kyiv School of Art. Labelled 'Russian' across most institutional records, with no reference to her Ukrainian formation.

Vera Pagava. Born in Tbilisi, Georgia. Labelled 'Empire Russe' in archival records — a framing that subsumes and erases Georgian identity entirely.

David Kakabadze. Born in Kutaisi/Tbilisi, Georgia. Absorbed into the Société Anonyme 'Russian avant-garde' canon without acknowledgement of Georgian nationality.

Sergei Parajanov. Soviet-Armenian filmmaker, Georgian-born. Birth name: Sarkis Hovsepi Parajanov. Labelled 'Soviet' in most records, with both national and ethnic identity omitted.

3. Propagation Chain

Misattribution propagates through a hierarchical chain of institutions. An absence or error in a Library of Congress authority record is adopted as authoritative by downstream institutions, which in turn supply training data to AI language models. The models then reproduce and amplify the original error at scale.

Tier	Institution / System	Role in Chain
Tier 0	Library of Congress	Authority records — original source
Tier 1	WorldCat / OCLC	Aggregation and redistribution
Tier 2	National libraries (Trove/NLA); Museum databases (MoMA, NGA, NGV)	Institutional adoption of authority labels
Tier 3	AI language models (Claude, GPT-4o, LLAMA); Web knowledge graphs (Google Knowledge)	Reproduction and amplification of error

4. AI Detection System

The detection tool queries AI language models about an artist's nationality and compares the response against the Record of Truth derived from the dataset.

4.1 Technical Implementation

Component	Detail
Language	Python 3.9+
Primary API	Anthropic Claude API (claude-opus-4-5)
Multi-model	Claude (Anthropic), GPT-4o (OpenAI API), LLAMA (Ollama, local)
Detection method	Keyword matching against correct and incorrect nationality terms
Image queries	Vision queries supported via artwork image URLs

4.2 Server Routes

The HTTP server is implemented using the Python standard-library HTTP server (no external frameworks). The following API routes are exposed:

Route	Method	Description
/api/query	GET	Single model query with misattribution assessment
/api/compare	GET	All three models compared side by side
/api/detect	GET	Misattribution detection for a given image URL
/api/artworks	GET	Full dataset records (filterable by artist)
/api/stats	GET	Dataset summary statistics
/api/images	GET	Image manifest and status

4.3 Source Files

File	Purpose
server/server.py	Main HTTP server
server/multi_llm.py	Multi-model comparison (Claude + GPT-4o + LLAMA)
server/claude_detector.py	Standalone CLI detection tool
server/image_pipeline.py	Artwork image management

5. Exhibition System

The exhibition is designed for a 12-screen LED installation configured as a 4-column by 3-row grid. Each screen is a separate browser window displaying one panel of a continuous full-canvas animation rendered in p5.js.

5.1 Visual Language

The artwork is rendered as a cloud of 1,800 luminous particles, each positioned at a pixel sampled from the source artwork image. Nearby particles are connected by fine threads. The visual language

operates across two opposing states:

Truth state: particles correctly positioned; threads form the real image clearly; warm cream luminosity on pure black background.

Misattribution state: particles displaced by a noise force field; threads tangled; cold blue-white colouration indicating corrupted data.

5.2 Phase Cycle

The animation cycles through eight phases automatically, with a total duration of approximately 35 seconds per artist:

Phase	Duration	Description
1. PRISTINE	3.5 s	Particles coalesce from void. Artwork emerges as constellation of light.
2. LABEL APPLY	3.5 s	Thin archival rectangle descends over the image.
3. CORRUPT	4.5 s	Noise force field tears particles from correct positions. Threads tangle.
4. PROPAGATE	5.0 s	Particle streams cross screen boundaries. Archive network activates.
5. MACHINE GAZE	4.5 s	Cold scanner light sweeps. AI systems read the corrupted data.
6. INTERVENTION	3.5 s	Warm light sweeps in. Spring force reclaims particles.
7. TRUTH	6.0 s	Particles lock to correct positions. Threads form the real image clearly.
8. FADE	2.0 s	Dissolves. Next artist begins.

5.3 Screen Synchronisation

Screen synchronisation is achieved via the browser LocalStorage API. Screen 1 (the leader) writes phase state every 500 milliseconds. All other screens read the shared state and align their animation phase accordingly. No server-side coordination is required.

5.4 Screen URLs

Each screen is loaded with URL parameters specifying its position in the grid (screen index, column count, row count):

Screen	URL
Screen 1	http://localhost:8000/index.html?screen=1&cols=4&rows=3
Screen 2	http://localhost:8000/index.html?screen=2&cols=4&rows=3
Screen 3	http://localhost:8000/index.html?screen=3&cols=4&rows=3
Screen 4	http://localhost:8000/index.html?screen=4&cols=4&rows=3
Screen 5	http://localhost:8000/index.html?screen=5&cols=4&rows=3
Screen 6	http://localhost:8000/index.html?screen=6&cols=4&rows=3
Screen 7	http://localhost:8000/index.html?screen=7&cols=4&rows=3

Screen 8	<code>http://localhost:8000/index.html?screen=8&cols=4&rows=3</code>
Screen 9	<code>http://localhost:8000/index.html?screen=9&cols=4&rows=3</code>
Screen 10	<code>http://localhost:8000/index.html?screen=10&cols=4&rows=3</code>
Screen 11	<code>http://localhost:8000/index.html?screen=11&cols=4&rows=3</code>
Screen 12	<code>http://localhost:8000/index.html?screen=12&cols=4&rows=3</code>
Single screen preview	<code>http://localhost:8000/index.html</code>
Curator controller	<code>http://localhost:8000/exhibition.html</code>

6. Visitor Kiosk

The kiosk interface (kiosk.html) is a visitor touch-screen component. Visitors select an artist; the system queries three AI models in parallel and displays their responses alongside the Record of Truth derived from the dataset.

URL: `http://localhost:8000/kiosk.html`

6.1 Layout

The kiosk displays four columns: Claude | GPT-4o | LLAMA | Record of Truth. Each AI response is revealed via a typewriter animation. A consensus bar indicates the degree of agreement or disagreement between the three models. Artist selector buttons allow navigation between Malevich, Exter, Pagava, Kakabadze, and Parajanov.

6.2 Operating Modes

Mode	Requirement	Behaviour
Demo mode	None (no API keys required)	Operates using pre-written responses that demonstrate typical misattribution
Live mode	ANTHROPIC_API_KEY environment variable	Queries Claude API in real time; GPT-4o and LLAMA also require their resp

7. Technical Requirements

7.1 Dependencies

Requirement	Detail
Python	3.9 or higher
Python packages	anthropic, openai, openpyxl, gdown
Environment variable	ANTHROPIC_API_KEY (required for live AI detection)
Browser	Chrome or Firefox (recommended)

7.2 Installation and Start

Install dependencies:

```
pip install anthropic openai openpyxl gdown
```

Start the server:

```
export ANTHROPIC_API_KEY="your-key-here"
```

```
python3 server/server.py
```

7.3 Repository

Field	Value
Repository	github.com/yamanidzet/Teona-Yamanidze
Branch	claude/analyze-art-datasets-fvjZt

8. Visual References

The visual aesthetic of the exhibition draws from the following precedents and principles:

Memo Akten, The Networked Condition — particle and thread systems that form recognisable figures on a black ground.

Mario Klingemann — painterly dissolution of image into data; the boundary between representation and information.

Pure black background with luminous warm-cream threads (clean / truth state); cold blue-white threads (corrupted / misattributed state).

Motion blur trail achieved via semi-transparent frame accumulation, lending each frame a painterly depth consistent with the archival subject matter.